

# Final

## Example 3.2

Calculate the inductance per meter of each phase of a three-phase transmission line (Figure E3.2). Assume that



Figure E3.2

1. Conductors are equally spaced,  $D$ , and have equal radii  $r$ .
2.  $i_a + i_b + i_c = 0$ .

**Solution** Using (3.21) for phase  $a$ , we have

$$\begin{aligned}\lambda_a &= \frac{\mu_0}{2\pi} \left( i_a \ln \frac{1}{r'} + i_b \ln \frac{1}{D} + i_c \ln \frac{1}{D} \right) \\ &= \frac{\mu_0}{2\pi} \left( i_a \ln \frac{1}{r'} - i_a \ln \frac{1}{D} \right) \\ &= \frac{\mu_0}{2\pi} i_a \ln \frac{D}{r'}\end{aligned}$$

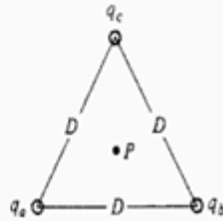
Thus,

$$l_a = \frac{\lambda_a}{i_a} = \frac{\mu_0}{2\pi} \ln \frac{D}{r'} = 2 \times 10^{-7} \ln \frac{D}{r'} \quad (3.23)$$

In the derivation use has been made of the equilateral spacing and the fact that  $i_a = -(i_b + i_c)$ . Because of symmetry we get the same results for  $\lambda_b$  and  $\lambda_c$ . Thus, we get the self-inductances  $l_a = l_b = l_c = (\mu_0/2\pi) \ln (D/r')$  henrys per meter. We also note that  $\lambda_a$  depends on  $i_a$  alone (i.e., there are no mutual inductance terms in this particular description). The same is true for  $b$  and  $c$  phases.

**Example 3.8**

Calculate an expression for the capacitance per meter of a three-phase transmission line [Figure E3.8(a)]. Assume that

**Figure E3.8(a)**

1. Conductors are equally spaced,  $D$ , and have equal radii  $r$ .
2.  $q_a + q_b + q_c = 0$ .

**Solution** Using  $a, b, c$  notation instead of 1, 2, 3, we find  $v_a$  by using (3.56):

$$\begin{aligned} v_a &= \frac{1}{2\pi\epsilon} \left( q_a \ln \frac{1}{r} + q_b \ln \frac{1}{D} + q_c \ln \frac{1}{D} \right) \\ &= \frac{q_a}{2\pi\epsilon} \ln \frac{D}{r} \end{aligned} \quad (3.60)$$

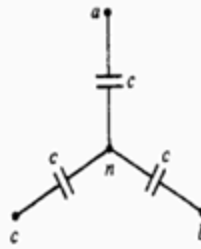
where we have used the assumption  $q_a + q_b + q_c = 0$ . In the same way we find identical relationships between  $v_b$  and  $q_b$ , and between  $v_c$  and  $q_c$ . We are now ready to calculate the three capacitances. To interpret the result physically, however, we digress briefly. Consider the potential of a point  $p$  equidistant from the conductors  $a, b$ , and  $c$ . Using (3.54) gives us

$$v_p = \frac{1}{2\pi\epsilon} \left( q_a \ln \frac{1}{R_{pa}} + q_b \ln \frac{1}{R_{pb}} + q_c \ln \frac{1}{R_{pc}} \right) = 0$$

Thus the point  $p$  is at the same potential as the point at infinity and may be taken as the datum point. In fact, construct an imaginary geometric line parallel to the conductors passing through the point  $P$ . Every point on this line is at the datum potential. Picking this also to be the neutral potential, we get the relationships between phase-to-neutral voltages and phase charges via three equal phase-neutral (distributed) capacitances:

$$c_a = c_b = c_c = c = \frac{2\pi\epsilon}{\ln(D/r)} \quad \text{F/m to neutral} \quad (3.61)$$

arranged as shown in Figure E3.8(b). This symmetric circuit may also be taken to be a representation of the physical cross section of the transmission line.



**Figure E3.8(b)**

In the more general case of unequal interphase spacings and bundled subconductors as described in Section 3.4, we assume that the lines are transposed and get

$$\bar{c}_a = \bar{c}_b = \bar{c}_c = \bar{c} = \frac{2\pi\epsilon}{\ln(D_m/R_b^c)} \quad \text{F/m to neutral} \quad (3.62)$$

where  $D_m$  is the GMD between phases and

$$R_b^c \triangleq (rd_{12} \cdots d_{1b})^{1/b} \quad b \geq 2 \quad (3.63)$$

We note that the only difference between (3.63) and (3.28) is that  $r$  is used instead of  $r'$ . If  $b = 1$ , it is convenient to define  $R_b^c = r$  so that we can use (3.62) to cover the non-bundled case.

**Example 4.1**

A 60-Hz 138-kV 3 $\phi$  transmission line is 225 mi long. The distributed line parameters are

$$r = 0.169 \, \Omega/\text{mi} \quad l = 2.093 \, \text{mH}/\text{mi} \quad c = 0.01427 \, \mu\text{F}/\text{mi} \quad g = 0$$

The transmission line delivers 40 MW at 132 kV with 95% power factor lagging. Find the sending-end voltage and current. Find the transmission-line efficiency.

**Solution** We first find  $z$  and  $y$  for  $\omega = 2\pi \cdot 60$ .

$$z = 0.169 + j0.789 = 0.807 \angle 77.9^\circ \, \Omega/\text{mi}$$

$$y = j5.38 \times 10^{-6} = 5.38 \times 10^{-6} \angle 90^\circ \, \text{mho}/\text{mi}$$

We then calculate  $Z_c$  and  $\gamma\ell$ .

$$Z_c = \sqrt{\frac{z}{y}} = 387.3 \angle -6.05^\circ \, \Omega$$

$$\gamma\ell = 225 \sqrt{zy} = 0.4688 \angle 83.95^\circ = 0.0494 + j0.466$$

In (4.10) we need  $\sinh \gamma\ell$  and  $\cosh \gamma\ell$ . We calculate these quantities next.

$$\begin{aligned} 2 \sinh \gamma\ell &\triangleq e^{\gamma\ell} - e^{-\gamma\ell} = e^{0.0494} e^{j0.466} - e^{-0.0494} e^{-j0.466} \\ &= 1.051 \angle 0.466 \, \text{rad} - 0.952 \angle -0.466 \, \text{rad} \end{aligned}$$

Thus

$$\sinh \gamma\ell = 0.452 \angle 84.4^\circ$$

In the preceding calculation it is important to note that in the formula  $e^{j\theta} = \cos \theta + j \sin \theta = 1 \angle \theta$ ,  $\theta$  is in radians. Thus  $e^{j0.466} = 1 \angle 0.466 \, \text{rad}$ . Continuing the calculation gives us

$$2 \cosh \gamma\ell = e^{\gamma\ell} + e^{-\gamma\ell} = 1.790 \angle 1.42^\circ$$

$$\cosh \gamma\ell = 0.8950 \angle 1.42^\circ$$

We now have all the transmission-line quantities needed in (4.10). It remains to find  $V_2$  and  $I_2$ . The quantities in the statement of the problem refer to  $3\phi$  and line-line quantities. Thus

$$|V_2| = 132 \times 10^3 / \sqrt{3} = 76.2 \text{ kV}$$

It is convenient to pick  $\angle V_2 = 0$ , in which case we get  $V_2 = 76.2 \angle 0^\circ \text{ kV}$ . We next find the per phase power supplied to the load.

$$P_{\text{load}} = \frac{40}{3} = 13.33 \text{ MW}$$

Since we are also given the power factor (0.95 lagging), we can find  $I_2$ . We have

$$P_{\text{load}} = 0.95 |V_2| |I_2|$$

Thus  $|I_2| = 184.1$ . Since  $I_2$  lags  $V_2$  by  $\cos^{-1} 0.95 = 18.195^\circ$ ,

$$I_2 = 184.1 \angle -18.195^\circ$$

Finally, plugging all these values into (4.10) gives us

$$\begin{aligned} V_1 &= V_2 \cosh \gamma \ell + Z_c I_2 \sinh \gamma \ell \\ &= 76.2 \times 10^3 \times 0.8950 \angle 1.42^\circ + 387.3 \angle -6.05^\circ \\ &\quad \times 184.1 \angle -18.195^\circ + 0.452 \angle 84.4^\circ \\ &= 68.20 \times 10^3 \angle 1.42^\circ + 32.23 \times 10^3 \angle 60.155^\circ \\ &= 89.28 \angle 19.39^\circ \text{ kV} \end{aligned}$$

Multiplying by  $\sqrt{3}$ , we find that the sending-end line-line voltage has magnitude 154.64 kV. We next use (4.10) to find  $I_1$ .

$$\begin{aligned} I_1 &= I_2 \cosh \gamma \ell + \frac{V_2}{Z_c} \sinh \gamma \ell \\ &= 162.42 \angle 14.76^\circ \end{aligned}$$

Thus, the magnitude of the sending-end current is 162.42 A. We now can calculate the efficiency of transmission. Since the per phase output power is 13.33 MW while the corresponding input power is

$$\begin{aligned} P_{12} &= \operatorname{Re} V_1 I_1^* = 89.28 \times 10^3 \times 162.42 \cos (19.39^\circ - 14.76^\circ) \\ &= 14.45 \text{ MW} \end{aligned}$$

we have

$$\eta = \frac{13.33}{14.45} = 0.92$$

That is, the efficiency of transmission is 92%.

### Example 10.1

In Figure E10.1(a) we are given  $V_1$  and  $S_2$  and are asked to find  $S_1$  and  $V_2$ . We consider the solution as a function of  $P_{D2}$  for  $P_{D2} \geq 0$ .

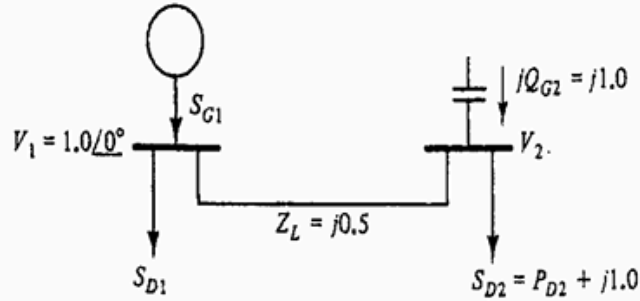


Figure E10.1(a)

**Solution** The capacitor in this case injects a specified power, while the voltage is not controlled. Thus bus 2 is a  $P, Q$  bus. In fact,

$$S_2 = S_{G2} - S_{D2} = -P_{D2}$$

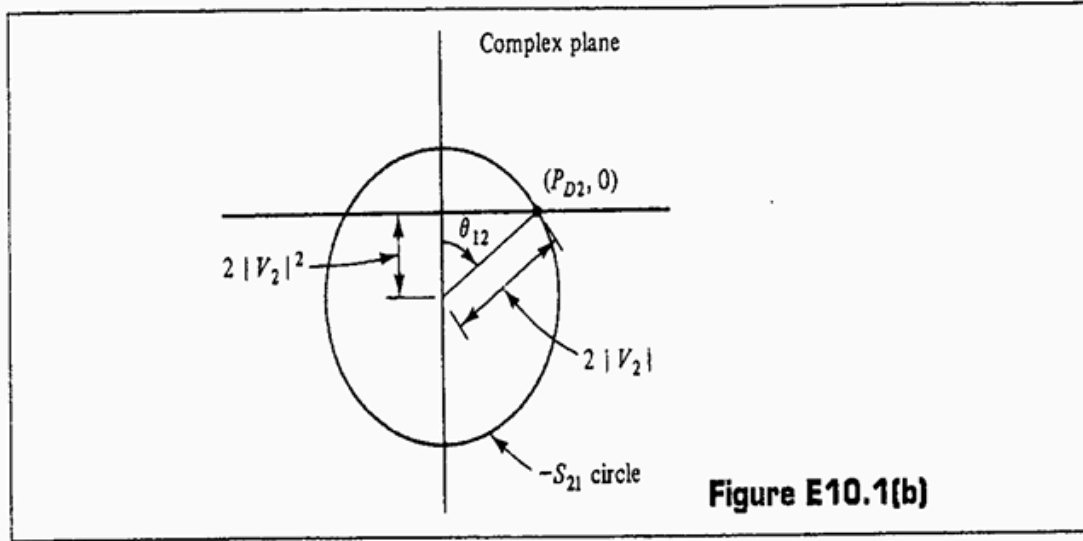
The easiest way to proceed is to use the same approach as in Example 4.8. Noting that  $S_2 = S_{21}$ , we draw a receiving-end circle [Figure E10.1(b)]. From the geometry of the figure we can see that if there is a solution,  $|V_2|$  must satisfy

$$4|V_2|^4 - 4|V_2|^2 + P_{D2}^2 = 0$$

which implies that

$$|V_2|^2 = \frac{1}{2}(1 \pm \sqrt{1 - P_{D2}^2})$$

Thus if  $P_{D2} > 1$ , there are no real solutions, and since  $|V_2|$  is necessarily real, there are no solutions. If  $P_{D2} = 1$ ,  $|V_2| = 1/\sqrt{2} = 0.707$ . If  $0 \leq P_{D2} < 1$ , there are *two* real solutions. In the cases where there *are* solutions, we may determine  $\angle V_2 = -\theta_{12}$  from the figure and  $S_1 = S_{12}$  from (10.4) or (4.29).



### Single line-to-ground faults currents

$$(13.5) [I_{af} \ I_{bf} \ I_{cf}] = A [I_{af}^0 \ I_{af}^+ \ I_{af}^-], \quad (13.6) A = [1 \ 1 \ 1; 1 \ \alpha^2 \ \alpha; 1 \ \alpha \ \alpha^2]$$

$$\text{Fault current: } I^f = [I_{af} \ I_{bf} \ I_{cf}] = [I^f \ 0 \ 0], \quad I_{af} = I_{af}^0 + I_{af}^+ + I_{af}^- = I^f,$$

$$I_{bf} = I_{bf}^0 + I_{bf}^+ + I_{bf}^- = (I^f/3)[1 \ \alpha^2 \ \alpha] = 0, \quad I_{cf} = I_{cf}^0 + I_{cf}^+ + I_{cf}^- = (I^f/3)[1 \ \alpha \ \alpha^2] = 0$$

$$A^{-1} = (1/3) [1 \ 1 \ 1; 1 \ \alpha \ \alpha^2; 1 \ \alpha^2 \ \alpha],$$

$$[I_{af}^0 \ I_{af}^+ \ I_{af}^-] = A^{-1} [I^f \ 0 \ 0] \\ = (I^f/3) [1 \ 1 \ 1]$$

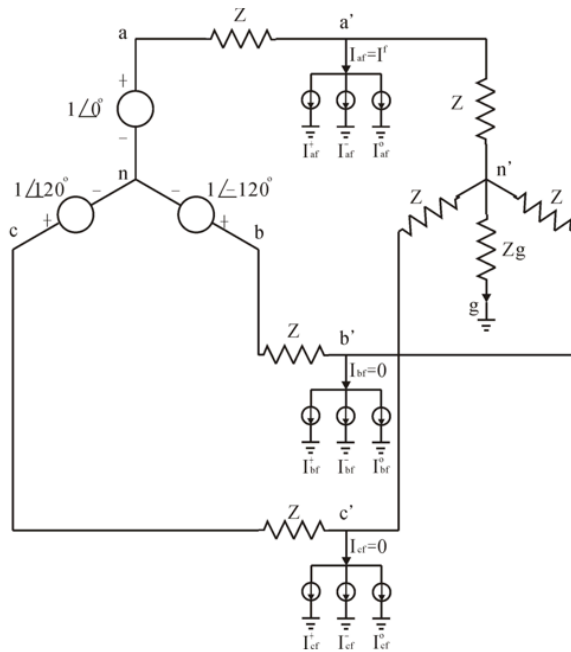
$$I_{af}^+ = I_{af}^- = I_{af}^0 = (I^f/3)$$

Using Superposition:

Positive Sequence

Negative Sequence

Zero Sequence



## Single line-to-ground faults currents: Positive Sequence

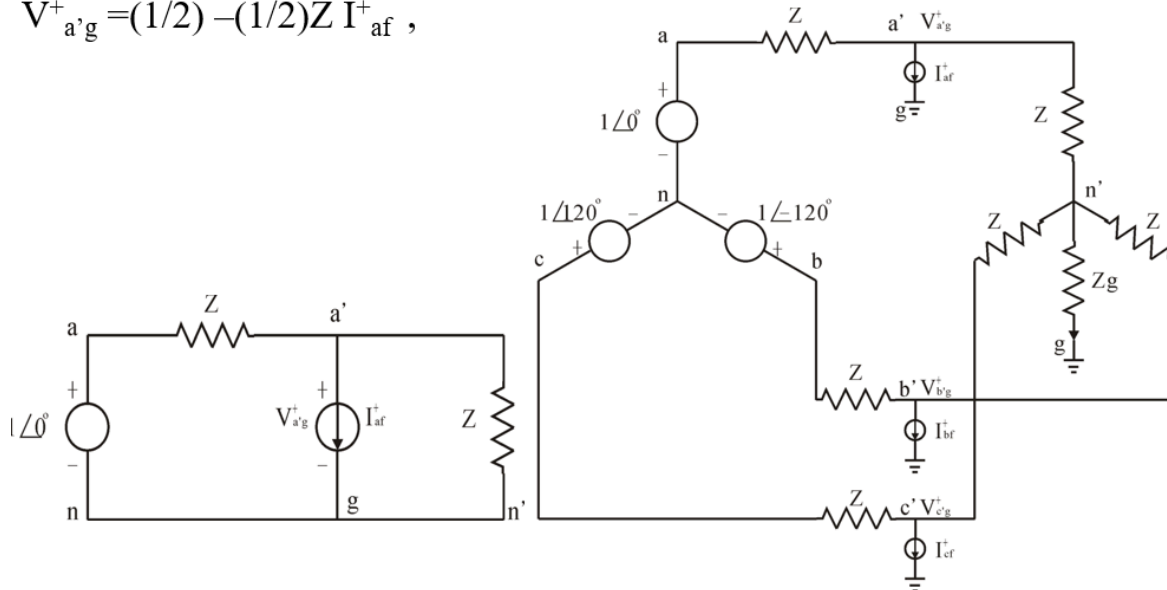
$$A^{-1} = (1/3) [1 \ 1 \ 1; 1 \ \alpha \ \alpha^2; 1 \ \alpha^2 \ \alpha],$$

$$[I_{af}^0 \ I_{af}^+ \ I_{af}^-] = A^{-1} [I^f \ 0 \ 0] = (I^f/3)[1 \ 1 \ 1]$$

$$I_{af}^+ = I_{af}^- = I_{af}^0 = (I^f/3)$$

Using Superposition:

$$V_{a'g}^+ = (1/2) - (1/2)Z I_{af}^+,$$



### Single line-to-ground faults currents: Negative Sequence

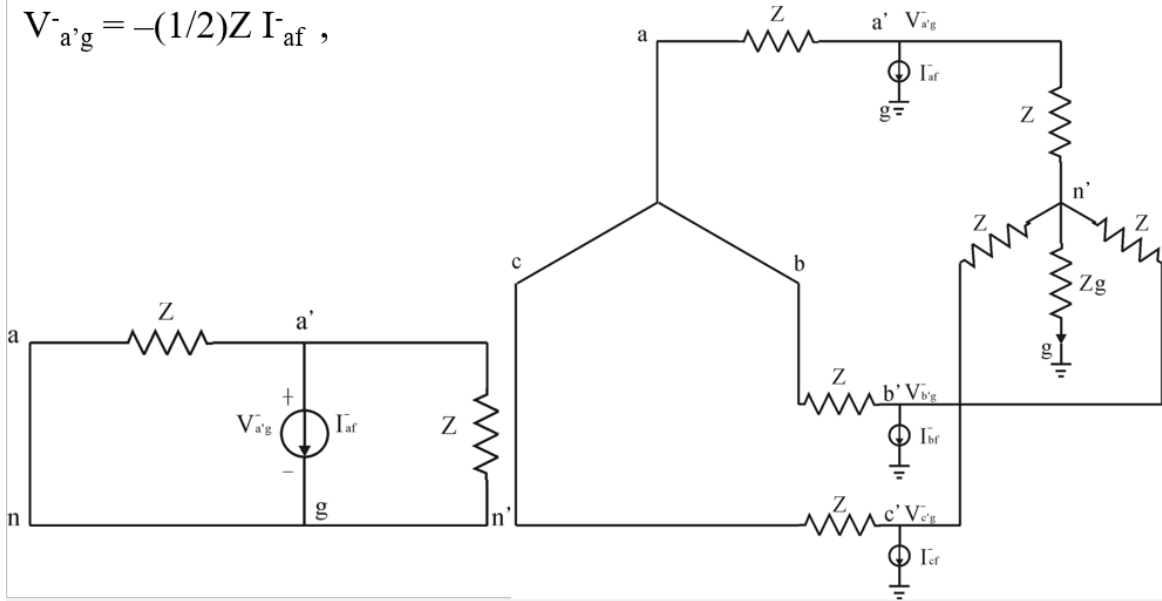
$$A^{-1} = (1/3) [1 \ 1 \ 1; 1 \ \alpha \ \alpha^2; 1 \ \alpha^2 \ \alpha] ,$$

$$[I_{af}^0 \ I_{af}^+ \ I_{af}^-] = A^{-1} [I^f \ 0 \ 0] = (I^f/3)[1 \ 1 \ 1]$$

$$I_{af}^+ = I_{af}^- = I_{af}^0 = (I^f/3)$$

Using Superposition:

$$V_{a',g}^- = -(1/2)Z I_{af}^- ,$$



### Single line-to-ground faults currents: Zero Sequence

(For n)  $I_a^0 = I_b^0 = I_c^0$ , and  $I_a^0 + I_b^0 + I_c^0 = 0$ ,  $\Rightarrow I_a^0 = I_b^0 = I_c^0 = 0$

(For n')  $I_a^0 = I_b^0 = I_c^0$ ,

$A^{-1} = (1/3) [1 \ 1 \ 1; 1 \ \alpha \ \alpha^2; 1 \ \alpha^2 \ \alpha]$ ,

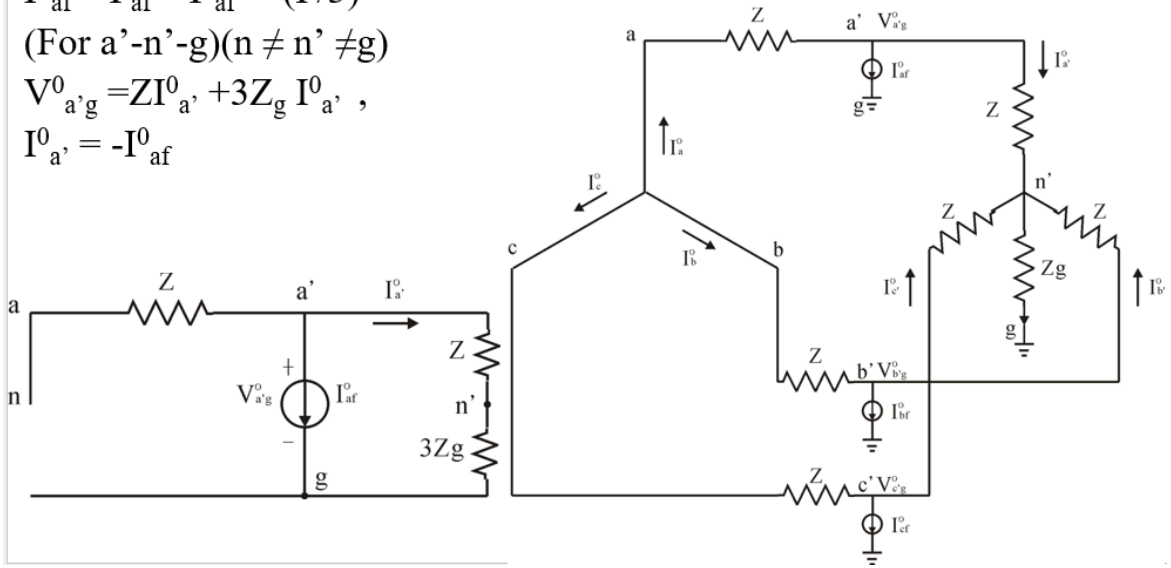
$[I_{af}^0 \ I_{af}^+ \ I_{af}^-] = A^{-1} [I^f \ 0 \ 0] = (I^f/3)[1 \ 1 \ 1]$

$I_{af}^+ = I_{af}^- = I_{af}^0 = (I^f/3)$

(For a'-n'-g) ( $n \neq n' \neq g$ )

$V_{a',g}^0 = Z I_a^0 + 3Z_g I_a^0$ ,

$I_a^0 = -I_{af}^0$



### 三相轉靜止座標(abc Reference Frame to Stationary Reference Frame)

$$[v_{qs} \ v_{ds}] = [2/3 \ -1/3 \ -1/3; 0 \ -1/\sqrt{3} \ 1/\sqrt{3}][v_a \ v_b \ v_c]$$

$$[v_a \ v_b \ v_c] = [1 \ 0; -1/2 \ -\sqrt{3}/2; -1/2 \ \sqrt{3}/2][v_{qs} \ v_{ds}]$$

3-phase 220V<sub>rms</sub> (60Hz, line-to-line),  $\omega = 2\pi f = 120\pi = 377 \text{ rad/sec}$ .

$v_a + v_b + v_c = 0$ , balanced system, ( $v_a + v_b + v_c \neq 0$ , unbalanced system)

$$v_a = 180 \cos \omega t, \ v_b = 180 \cos (\omega t - 120^\circ), \ v_c = 180 \cos (\omega t + 120^\circ),$$

#### abc Reference Frame to Stationary Reference Frame

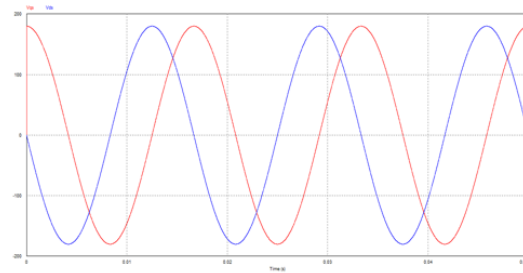
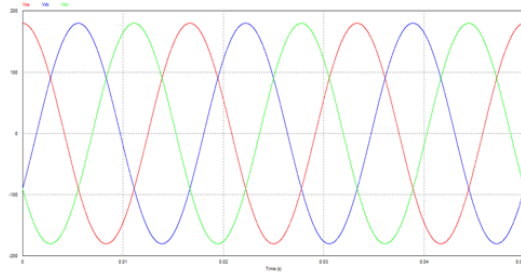
$$v_{qs} = (2/3)v_a + (-1/3)v_b + (-1/3)v_c = (2/3)v_a + (1/3)v_a = v_a = 180 \cos \omega t$$

$$v_{ds} = (-1/\sqrt{3})v_b + (1/\sqrt{3})v_c = (-1/\sqrt{3})[v_a + 2v_b] = -180 \sin \omega t$$

#### Stationary Reference Frame to abc Reference Frame

$$v_a = v_{qs}, \ v_b = (-1/2)v_{qs} + (-\sqrt{3}/2)v_{ds} = 180 \cos(\omega t - 120^\circ),$$

$$v_c = (-1/2)v_{qs} + (\sqrt{3}/2)v_{ds} = 180 \cos(\omega t + 120^\circ)$$



## 靜止座標轉同步座標(Stationary Reference Frame to Synchronous Reference Frame)

$$[v_{qe} \ v_{de}] = [\cos \omega t \ -\sin \omega t; \sin \omega t \ \cos \omega t][v_{qs} \ v_{ds}]$$

$$[v_{qs} \ v_{ds}] = [\cos \omega t \ \sin \omega t; -\sin \omega t \ \cos \omega t][v_{qe} \ v_{de}]$$

3-phase 220V<sub>rms</sub> (60Hz, line-to-line),  $\omega = 2\pi f = 120\pi = 377$  rad/sec.

$$v_{qs} = 180 \cos \omega t, \ v_{ds} = -180 \sin \omega t,$$

### Stationary Reference Frame to Synchronous Reference Frame

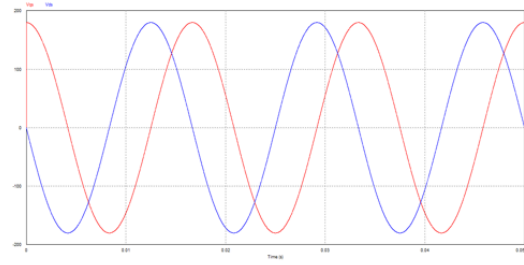
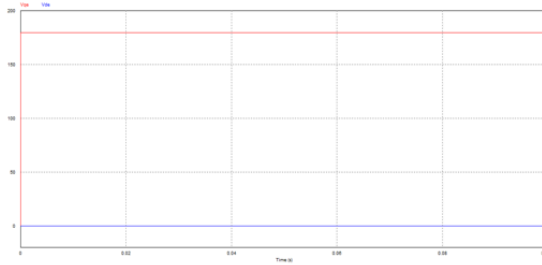
$$v_{qe} = \cos \omega t \ v_{qs} - \sin \omega t \ v_{ds} = 180$$

$$v_{de} = \sin \omega t \ v_{qs} + \cos \omega t \ v_{ds} = 0$$

### Synchronous Reference Frame to Stationary Reference Frame

$$v_{qs} = \cos \omega t \ v_{qe} + \sin \omega t \ v_{de} = 180 \cos \omega t,$$

$$v_{ds} = -\sin \omega t \ v_{qe} + \cos \omega t \ v_{de} = -180 \sin \omega t$$



靜止座標轉非同步座標(Stationary Reference Frame to Asynchronous Reference Frame)( $\omega_1 \neq \omega$ , assume 59Hz,  $\omega_1 = 118\pi$  rad/sec.)

$$[v_{qe} \ v_{de}] = [\cos \omega_1 t \ -\sin \omega_1 t; \sin \omega_1 t \ \cos \omega_1 t][v_{qs} \ v_{ds}]$$

$$[v_{qs} \ v_{ds}] = [\cos \omega_1 t \ \sin \omega_1 t; -\sin \omega_1 t \ \cos \omega_1 t][v_{qe} \ v_{de}]$$

3-phase 220V<sub>rms</sub> (60Hz, line-to-line) ,  $\omega = 2\pi f = 120\pi = 377$  rad/sec.

$$v_{qs} = 180 \cos \omega t, \ v_{ds} = -180 \sin \omega t,$$

**Stationary Reference Frame to Asynchronous Reference Frame**

$$v_{qe} = \cos \omega_1 t v_{qs} - \sin \omega_1 t v_{ds} = 180 \cos (\omega_1 - \omega)t \neq 180$$

$$v_{de} = \sin \omega_1 t v_{qs} + \cos \omega_1 t v_{ds} = 180 \sin (\omega_1 - \omega)t \neq 0$$

**Asynchronous Reference Frame to Stationary Reference Frame**

$$v_{qs} = \cos \omega_1 t v_{qe} + \sin \omega_1 t v_{de} = v_{qs} = 180 \cos \omega t,$$

$$v_{ds} = -\sin \omega_1 t v_{qe} + \cos \omega_1 t v_{de} = v_{ds} = -180 \sin \omega t$$

